

Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe [Metric]

AASHTO Designation: M 170M-04¹

ASTM Designation: C 76M-02



1. SCOPE

1.1. This specification covers reinforced concrete pipe intended to be used for the conveyance of sewage, industrial wastes, and storm water, and for the construction of culverts.

1.2. This specification is the metric counterpart of M 170.

Note 1—This specification is a manufacturing and purchase specification only, and does not include requirements for bedding, backfill, or the relationship between field load condition and the strength classification of pipe. However, experience has shown that the successful performance of this product depends upon the proper selection of the class of pipe, type of bedding and backfill, controlled manufacture in the plant, and care that installation conforms to the construction specifications. The owner of the reinforced concrete pipe specified herein is cautioned that he must correlate the field requirements with the class of pipe specified and provide inspection at the construction site.

Note 2—Attention is called to the specification for reinforced concrete D-load culvert, storm drain, and sewer pipe (M 22.2M).

2. REFERENCED DOCUMENTS

2.1. *AASHTO Standards:*

- M 6, Fine Aggregate for Portland Cement Concrete
- M 31M/M 31, Deformed and Plain Billet-Steel Bars for Concrete Reinforcement
- R M 32WH 32, Steel Wire, Plain, for Concrete Reinforcement
- M 55M/M 55, Steel Welded Wire Reinforcement, Plain, for Concrete
- R M C0, Coarse Aggregate for Portland Cement Concrete
- n M 85, Portland Cement
- m M 86M, Concrete Sewer, Storm Drain, and Culvert Pipe [Metric]
- n M 148, Liquid Membrane-Forming Compounds for Curing Concrete
- o M 221M/M 221, Steel Welded Wire Reinforcement, Deformed, for Concrete
- M 225M/M 225, Steel Wire, Deformed, for Concrete Reinforcement
- M 240, Blended Hydraulic Cement
- n M 242M, Reinforced Concrete D-Load Culvert, Storm Drain, and Sewer Pipe [Metric]
- n M 262, Concrete Pipe and Related Products
- n M 295, Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture

- 2.2.
- ASTM Siandar<Lr.

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C 1116, Specification for Fiber-ReinforcedConcrctc• and Shotcrete

3.

TERMINOLOGY

- 3.1.
- NeJriifio zr—For definitions of terms relating to concrete pipe, see M 262.

4.

CLASSIFICATION

- 4.1.
- f*ibe man«factured in accordance with this spcciFicaiiun stial Lo° ot“ IM't C)(1S5”c”k idcntificd as Chess I, Class II, Class III, Class EV, and Class V. The corres inning stret gtli rcxjuizementsare prescribed in Table's 1 to S.**

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Table 1—Design Requirements for Class I Reinforced Concrete Pipe

Internal Diameter, D, mm	Reinforcement, cm ² /Linear m of Pipe Wall								
	Wall A					Wall B			
	Concrete Strength, 27.6 MPa					Concrete Strength, 27.6 MPa			
	Wall Thickness, mm	Circular Reinforcement ¹			Elliptical Reinforcement ¹	Wall Thickness, mm	Circular Reinforcement ¹		
		Inner Cage	Outer Cage				Inner Cage	Outer Cage	Elliptical Reinforcement ¹
1500	125	5.3	1.2		5.9	119	4.4	2.6	4.9
1650	138	6.4	3.8		11.1	163	1.1	1.4	5.9
1800	150	7.4	4.4		8.3	175	6.1	3.7	6.8
1950	163	8.5	5.1		10.6	188	6.8	4.1	7.6
2100	175	9.5	5.7		10.6	201	7.8	4.7	8.1
2250	188	10.4	6.2		11.1	213	8.1	5.2	9.7
2400	200	11.4	6.8		12.7	225	9.7	5.9	10.4
Concrete Strength, 34.5 MPa									
2250	213	13.3	8.0	Inner Circular Plus Elliptical	53 to 60	238	11.4	6.8	Inner Circular Plus Elliptical
2700	225	14.4	8.6	Inner Circular Plus Elliptical	5.8 to 8.6	250	12.9	7.7	Inner Circular Plus Elliptical
2850	—	—	—	—	—	—	—	—	—
3000	—	—	—	—	—	—	—	—	—
3150	—	—	—	—	—	—	—	—	—
3300	—	—	—	—	—	—	—	—	—
3450	—	—	—	—	—	—	—	—	—

For special design, see Section 7.2 or with the permission of the owner utilize the provision of M 22h1. Shells may be incorporated but not those shown for variations in diameter, loading or wall thickness. Pipe over 2400 mm in diameter shall have two circular cages or an inner circular cage plus one elliptical cage.

As an alternative to design requiring separate inner and outer circular reinforcement, the reinforcement may be positioned to include the following:
(1) An inner circular cage plus an elliptical cage such that arcs of the elliptical cage shall not be less than the arc of the inner cage in the horizontal plane.
(2) An inner circular cage plus an elliptical cage such that the arcs of the inner cage shall not be less than the arcs of the elliptical cage in the horizontal plane.
(3) An inner and outer cage plus an elliptical cage in accordance with Figure 2.

- Reinforcement shall be placed in the pipe wall by means of chairs, or other positive means throughout the concrete casting operation.

Note: See Section 5 for basis of acceptance specified by the owner. The strength test requirements in newtons per linear meter of pipe under the three-edge-bearing method shall be as follows:
D-load (test) in newtons per linear meter per millimeter of diameter to produce the 0.1-mm crack in the D-lump to produce the 0.1-mm crack and the load is specified below, multiplied by the internal diameter of the pipe in millimeters.

D-load to produce 0.3-mm crack 10
D-load to produce in ultimate load 1.0

Table 2—Design Requirements for Class II Reinforced Concrete Pipe^a

Reinforcement, cm ² /linear m of Pipe 6' all														
Internal Designated Diameter, mm	Wa8 A				Wt11 O				Wali c					
	Concrete Strength, 27.6 MPa				Concrete Strength, 27.6 MPa				Concrete Strength, 27.6 MPa					
	Well Thickness, mm	Circular Reinforcement ^a		Elliptical Reinforcement ^c	y, g; Thickness, mm	Circular Reinforcement ^b		Elliptical Reinforcement ^c	Wall Thickness, mm	Circular Reinforcement ^b		Elliptical Reinforcement ^c		
		Inner Cage	Outer Cage			Inner Cage	Outer Cage			Inner Cage	Outer Cage			
300	44	1.5"	—	—	50	1.5"	—	—	69	1.5"	—	—		
430	50	1.5"	—	1.5"	63	1.5"	—	1.5"	82	1.5"	—	1.5"		
525	57	2.3	—	2.1	69	1.5"	—	1.5"	88	1.5"	—	1.5"		
600	63	2.8	—	2.3	76	1.5"	—	1.5"	94	1.5"	—	1.5"		
675	66	3.2	—	2.8	82	2.1	—	2.3	100	1.5"	—	1.5"		
750	72	3.6	—	3.0	88	2.1	—	2.3	106	1.5"	—	1.5"		
825	72	3.6	—	3.0	94	2.1	—	2.3	113	1.5"	—	1.5"		
900	75	3.6	1.6	3.2	100	2.1	1.5	2.8	119	1.5	1.5	1.7		
1050	69	3.4	2.0	3.8	113	3.2	1.9	3.6	132	2.1	1.5	2.3		
1200	100	4.5	2.7	4.9	125	3.8	2.3	4.2	144	3.0	1.5	3.2		
1350	113	5.3	3.2	5.9	141	4.7	2.8	5.1	157	3.6	2.2	3.8		
1500	125	6.4	3.8	7.0	151	5.3	3.2	5.9	169	4.7	2.8	4.1		
1650	136	7.4	4.4	8.3	163	6.6	3.1	7.2	182	5.3	3.2	3.9		
1800	150	8.7	5.2	9.5	175	7.4	4.4	8.1	200	6.4	3.5	7.0		
1950	161	9.7	5.5	10.1	188	8.5	5.1	9.3	207	7.4	4.4	8.3		
2100	175	10.6	6.5	12.1	200	9.7	5.8	10.3	219	8.7	5.5	9.7		
2250	183	12.1	7.3	13.3	213	10.1	6.5	11.1	232	10.2	6.5	11.1		
2400	200	13.1	7.9	14.5	225	12.1	7.3	12.1	244	11.6	7.0	12.9		
Concrete Strength, 34.5 MPa														
2450	213	16.1	9.7	Inner Circular Plus Elliptical	6 9.7	238	14.4	8.5	Inner Circular Plus Elliptical	5.8 6.6	257	13.1	7.9 Inner Circular Plus Elliptical	5.2 7.9
2700	225	18.0	10.1	Inner Circular Plus Elliptical	7.2 10.8	250	16.1	9.7	Inner Circular Plus Elliptical	6.4 9.7	269	14.8	9.9 Inner Circular Plus Elliptical	5.9 D9
330	—	—	—	—	—	—	—	—	—	—	—	—	—	
3450	—	—	—	—	—	—	—	—	—	—	—	—	—	

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		Elliptical and quadrant steel must be he(d in place by means of holding rods, chain. or uthcr po.siti•c means ihmut;h tbc c«:re ««'iny• nycrat or..
S	“	For these rlssscs and size, the miniofum practical ytccl reinforcement is specified, The acrucl ullimntc strength ix greater ihaij the minruum 9lrcnglh speci died for no=rc forccu pops of c«ui 'stent dlemctm in 86h1 Ae an alternative, single cage rcinfortcmcnl may be used. Nc reinforcement area in square ccniimclcrs per linear mctct shall he 4.2 for V•all II and ?.4 her \\\ell C Ante: Sec Section J for bssis of acceptance .rprcifed by the ovmc. The strength tcyt requirement in ncwtons per l'ncar mcicr nf p:ie unric• ihc thrcc•cdgc-bcanng method she'l hc cither ihc D-hoed (icsi load expressed in nowtons petlincxrmctcrptrmillimeczofdzm**) !opmducclhco3-mmVck o«hcDloads!oproducc+hc03.mn crack.and hculttm3tzloxdx>spcnñcdhcov,rnuhipWcdDythc acm*.dxmccrofthcppeinmiltimctc.
		D-hoed to producc a 0.3-mm crack 50.0 D-load to produce the ultimate load ?S.0

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Table Design Requirements for Class III Reinforced Concrete Pipe"

Wall A												
Concrete Strength, 27.6 MPa												
Designated Diameter, mm	Wall Thickness, mm	Circular			Wall Thickness, mm	Elliptical			Wall Thickness, mm	Elliptical		
		Reinforcement*				Reinforcement*				Reinforcement*		
		Inner	Outer	Elliptical		Inner	Outer	Elliptical		Inner	Outer	Elliptical
		Cage	Cage	Reinforcement†		Cage	Cage	Reinforcement†		Cage	Cage	Reinforcement†
		mm	mm	mm		mm	mm	mm		mm	mm	mm
300	44	1.5"	—	—	69	1.5"	—	—	82	1.5"	—	—
371	47	1.5"	—	—	75	1.5"	—	—	88	1.5"	—	—
450	50	1.5"	—	1.1°	82	1.5"	—	1.5"	94	1.5"	—	1.5"
525	57	1.5"	—	2.0	88	1.5"	—	1.5"	100	1.5"	—	1.5"
600	63	1.5"	—	3.0	94	1.5"	—	1.5"	107	1.5"	—	1.5"
675	69	1.5"	—	3.6	100	1.5"	—	1.5"	113	1.5"	—	1.5"
750	75	1.5"	—	3.8	107	1.5"	—	1.5"	119	1.5"	—	1.5"
B2S	82	1.5"	—	4.2	113	1.5"	—	1.5"	125	1.5"	—	1.5"
900	88	1.5"	—	4.7	119	1.5"	—	1.5"	132	1.5"	—	1.5"
M10	94	1.5"	—	5.0	125	1.5"	—	1.5"	144	1.5"	—	1.5"
	100	1.5"	—	5.4	132	1.5"	—	1.5"	157	1.5"	—	1.5"
				5.9	144	1.5"	—	1.5"	169	1.5"	—	1.5"
				6.4	157	1.5"	—	1.5"	182	1.5"	—	1.5"
				6.9	169	1.5"	—	1.5"	194	1.5"	—	1.5"
				7.4	175	1.5"	—	1.5"	207	1.5"	—	1.5"
				7.9					219	1.5"	—	1.5"
				8.4					232	1.5"	—	1.5"
				8.9					244	1.5"	—	1.5"
				9.4					257	1.5"	—	1.5"
				9.9					269	1.5"	—	1.5"
				10.4					282	1.5"	—	1.5"
				10.9					294	1.5"	—	1.5"
				11.4					307	1.5"	—	1.5"
				11.9					319	1.5"	—	1.5"
				12.4					332	1.5"	—	1.5"
				12.9					344	1.5"	—	1.5"
				13.4					357	1.5"	—	1.5"
				13.9					369	1.5"	—	1.5"
				14.4					382	1.5"	—	1.5"
				14.9					394	1.5"	—	1.5"
				15.4					407	1.5"	—	1.5"
				15.9					419	1.5"	—	1.5"
				16.4					432	1.5"	—	1.5"
				16.9					444	1.5"	—	1.5"
				17.4					457	1.5"	—	1.5"
				17.9					469	1.5"	—	1.5"
				18.4					482	1.5"	—	1.5"
				18.9					494	1.5"	—	1.5"
				19.4					507	1.5"	—	1.5"
				19.9					519	1.5"	—	1.5"
				20.4					532	1.5"	—	1.5"
				20.9					544	1.5"	—	1.5"
				21.4					557	1.5"	—	1.5"
				21.9					569	1.5"	—	1.5"
				22.4					582	1.5"	—	1.5"
				22.9					594	1.5"	—	1.5"
				23.4					607	1.5"	—	1.5"
				23.9					619	1.5"	—	1.5"
				24.4					632	1.5"	—	1.5"
				24.9					644	1.5"	—	1.5"
				25.4					657	1.5"	—	1.5"
				25.9					669	1.5"	—	1.5"
				26.4					682	1.5"	—	1.5"
				26.9					694	1.5"	—	1.5"
				27.4					707	1.5"	—	1.5"
				27.9					719	1.5"	—	1.5"
				28.4					732	1.5"	—	1.5"
				28.9					744	1.5"	—	1.5"
				29.4					757	1.5"	—	1.5"
				29.9					769	1.5"	—	1.5"
				30.4					782	1.5"	—	1.5"
				30.9					794	1.5"	—	1.5"
				31.4					807	1.5"	—	1.5"
				31.9					819	1.5"	—	1.5"
				32.4					832	1.5"	—	1.5"
				32.9					844	1.5"	—	1.5"
				33.4					857	1.5"	—	1.5"
				33.9					869	1.5"	—	1.5"
				34.4					882	1.5"	—	1.5"
				34.9					894	1.5"	—	1.5"
				35.4					907	1.5"	—	1.5"
				35.9					919	1.5"	—	1.5"
				36.4					932	1.5"	—	1.5"
				36.9					944	1.5"	—	1.5"
				37.4					957	1.5"	—	1.5"
				37.9					969	1.5"	—	1.5"
				38.4					982	1.5"	—	1.5"
				38.9					994	1.5"	—	1.5"
				39.4					1007	1.5"	—	1.5"
				39.9					1019	1.5"	—	1.5"
				40.4					1032	1.5"	—	1.5"
				40.9					1044	1.5"	—	1.5"
				41.4					1057	1.5"	—	1.5"
				41.9					1069	1.5"	—	1.5"
				42.4					1082	1.5"	—	1.5"
				42.9					1094	1.5"	—	1.5"
				43.4					1107	1.5"	—	1.5"
				43.9					1119	1.5"	—	1.5"
				44.4					1132	1.5"	—	1.5"
				44.9					1144	1.5"	—	1.5"
				45.4					1157	1.5"	—	1.5"
				45.9					1169	1.5"	—	1.5"
				46.4					1182	1.5"	—	1.5"
				46.9					1194	1.5"	—	1.5"
				47.4					1207	1.5"	—	1.5"
				47.9					1219	1.5"	—	1.5"
				48.4					1232	1.5"	—	1.5"
				48.9					1244	1.5"	—	1.5"
				49.4					1257	1.5"	—	1.5"
				49.9					1269	1.5"	—	1.5"
				50.4					1282	1.5"	—	1.5"
				50.9					1294	1.5"	—	1.5"
				51.4					1307	1.5"	—	1.5"
				51.9					1319	1.5"	—	1.5"
				52.4					1332	1.5"	—	1.5"
				52.9					1344	1.5"	—	1.5"
				53.4					1357	1.5"	—	1.5"
				53.9					1369	1.5"	—	1.5"
				54.4					1382	1.5"	—	1.5"
				54.9					1394	1.5"	—	1.5"
				55.4					1407	1.5"	—	1.5"
				55.9					1419	1.5"	—	1.5"
				56.4					1432	1.5"	—	1.5"
				56.9					1444	1.5"	—	1.5"
				57.4					1457	1.5"	—	1.5"
				57.9					1469	1.5"	—	1.5"

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Designated Diameter, mm	Wall Thickness, mm	Concrete Strength, 27.6 MPa							
		Internal Diameter, mm	Circular Reinforcement		Elliptical Reinforcement ^c	Thickness, mm	Elliptical Reinforcement ^c	Outer Cage	
			Inner Cage	Outer Cage					
2700	25.8	15.5	Inner	10.3	Wall A	Inner	9.2	12.6	
3000	"	"	"	"	"	"	"	"	
3150	"	"	"	"	"	"	"	"	
3600	"	"	"	"	"	"	"	"	

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As an alternative to designs requiring both inner and outer circular cages the reinforcement may be positioned and proportioned in either of the following manners: (1) An inner circular cage plus an elliptical cage.

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(2) An inner and outer cage plus quadrant mats in accordance with Figure 1; or (3) An inner and outer cage plus an elliptical cage. Elliptical and quadrant steel must be held in place by means of holding rods, chairs, or other positive means throughout the entire length of the cage. For these classes and sizes, the minimum practical steel reinforcement is specified. The actual ultimate strength is greater than the specified strength. As an alternative, single cage reinforcement may be used. The reinforcement area in square centimeters per linear meter shall be 6

Note: See Section 5 for basis of acceptance specified by the owner. The strength test requirements

Internal Designated Diameter, mm	Wall Thickness, mm	Wall A										
		Concrete Strength, 34.5 MPa					Concrete Strength, 27.6 MPa					
		Circular Reinforcement ^b			Thickness, mm	Circular Reinforcement ^b			Thickness, mm	Circular Reinforcement ^b		
		Inner Cage	Outer Cage	Elliptical Reinforcement ^c		Inner Cage	Outer Cage	Elliptical Reinforcement ^c		Inner Cage	Outer Cage	Elliptical Reinforcement ^c
300	44	3.2	—	—	50	1.5	—	—	69	1.5 ^d	—	—

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Table W-X. *oiifinued*

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maximal load per millimetre of diameter 't' to produce a 0.3 mm crack, or the load to produce the 0.3 mm crack, and the ultimate load capacity

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Notes: See Section 5 for basis of acceptance specified by the owner. The strength

M 170M-9

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Table 5—Design Requirements for Class V Reinforced Concrete Pipe^a

Internal Designed Diameter, mm	Reinforcement, cm ² /Linear m of Pipe Wall									
	Wall A					Wall B				
	Concrete Strength, 41.4 MPa					Concrete Strength, 41.4 MPa				
	Wall Thickness, mm	Circular Reinforcement ^b		Wall Thickness, mm	Elliptical Reinforcement ^c	Circular Reinforcement ^b		Wall Thickness, mm	Elliptical Reinforcement ^c	Concrete Strength, 41.4 MPa Wall C
		Inner Cage	Outer Cage			Inner Cage	Outer Cage			Concrete Strength, 41.4 MPa Wall C
300	•	—	—	50	—	2.1	—	69	—	1.5 ^d
375	•	—	—	57	—	3.0	—	75	—	1.5 ^d
450	•	—	—	63	—	4.0	—	82	—	—
525	•	—	—	69	—	5.1	—	88	—	—
600	•	—	—	75	—	6.4	—	94	—	2.8
675	•	—	—	82	—	8.0	4.8	100	—	3.4
750	•	—	—	88	—	8.7	5.2	107	—	4.2
825	•	—	—	94	—	9.7	5.8	113	—	5.3
900	•	—	—	100	—	10.6	6.4	119	—	6.3
1050	•	—	—	113	—	12.7	7.6	132	—	8.5
1200	•	—	—	125	—	15.5	9.3	144	—	11.0
1350	•	—	—	•	—	—	—	157	—	13.5
1500	•	—	—	•	—	—	—	169	—	16.5
1650	•	—	—	•	—	—	—	182	—	19.7
1800	•	—	—	•	—	—	—	194	—	23.3
1950	•	—	—	•	—	—	—	•	—	—
2100	•	—	—	•	—	—	—	•	—	—
2250	•	—	—	•	—	—	—	•	—	—
2400	•	—	—	•	—	—	—	•	—	—
2550	•	—	—	•	—	—	—	•	—	—
2700	•	—	—	•	—	—	—	•	—	—
2850	•	—	—	•	—	—	—	•	—	—
3000	•	—	—	•	—	—	—	•	—	—
3150	•	—	—	•	—	—	—	•	—	—
3300	•	—	—	•	—	—	—	•	—	—
3450	•	—	—	•	—	—	—	•	—	—
3600	•	—	—	•	—	—	—	•	—	—

^a For modified or special designs, see Section 7.2 or with the permission of the owner utilize the provision of M 242M. Steel areas may be interpolated between those shown for variations in diameter, loading, or wall thickness. Pipe over 2400 mm in diameter shall have two circular cages or an inner circular cage plus one elliptical cage.

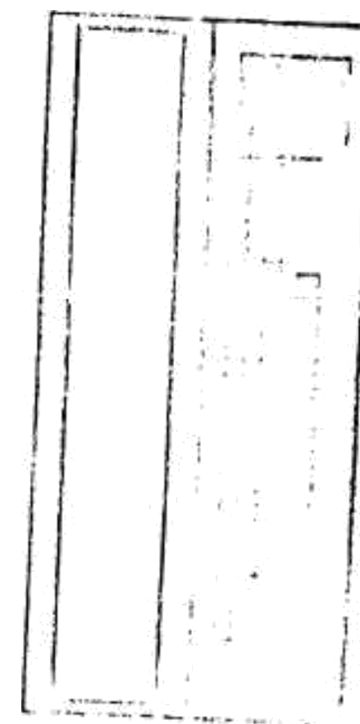
^b As an alternative to designs requiring both inner and outer circular cages the reinforcement may be positioned and proportioned in either of the following manners: (1) an inner circular cage plus an elliptical cage such that the area of elliptical cage shall not be less than that specified for the outer cage in the table and the total area of the inner circular cage plus the elliptical cage shall not be less than that specified for the inner cage in the table; (2) an inner and outer cage plus quadrant mats in accordance with Figure 1; or (3) an inner and outer cage plus an elliptical cage in accordance with Figure 2.

Continued on next page.

5—C

Elliptical and quadrant steel must be held in place by means of holding rods, chairs, or other positive means throughout the entire casting operation.

Notes: Section S for hadr of acoq tanc cpe « lñi bY thee•m•r. Th* ctr•nEh lest **W**!! meter of pipe under the three-edge-bearing method shall be either the D-load (test load expressed in newtons per 101 cnr md ef pw millim i er of ditfnetcr) to produce the 0.3-mm crsck• or the D-loads !
D- load la producec OJ c zsck \ 40.0
D• lox d la produce the ullimclc load 1 ISD



5. BASIS OF ACCEPTANCE

- 5.1. Unless otherwise specified by the owner at the time of, or before placing an order, two separate and alternative bases of acceptance are provided as follows:
- 5.1.1. *Acceptance on the Basis of Flexural Load-Bearing Tests, Material Tests, and Inspection of Manufactured Pipe for Visual Defects and Imperfections*—Acceptability of the pipe in all diameters and classes produced in accordance with Section 7.1 or Section 7.2 shall be determined by the results of the flexural load-bearing tests for either the load to produce a 0.3-mm crack, or at the option of the owner, the load to produce a 0.3-mm crack and the ultimate strength of the pipe; by such material tests as are required in Sections 6.1, 6.2, and 6.4; by absorption tests on selected samples of concrete from the wall of the pipe; and by visual inspection of the finished pipe to determine its conformance with the accepted design and its freedom from defects.
- 5.1.2. *Acceptance on the Basis of Material Tests and Inspection of Manufactured Pipe for Defects and Imperfections*—Acceptability of the pipe in all diameters and classes produced in accordance with Section 7.1 or Section 7.2 shall be determined by the results of such material tests as are required in Sections 6.1, 6.2, and 6.4; by crushing tests on concrete cores or cured concrete cylinders; by absorption tests on selected samples from the wall of the pipe; and by inspection of the finished pipe including amount and placement of reinforcement to determine its conformance with the accepted design and its freedom from defects.
- 5.1.3. When agreed upon by the owner and the manufacturer, any portion or any combination of the tests itemized in Section 5.1.1 or Section 5.1.2 may form the basis of acceptance.
- 5.2. *Age for Acceptance*—Pipe shall be considered ready for acceptance when it conforms to the requirements as indicated by the specified tests.

6. MATERIALS

- 6.1 *Reinforced Concrete*—The reinforced concrete shall consist of concrete, cementitious materials, mineral aggregates, and water, in which steel has been embedded in such a manner that the steel and concrete act together.
- 6.2. *Cementitious Materials*.
- 6.2.1. *Cement*—Cement shall conform to the requirements for portland cement of M 85, or shall be portland blast-furnace slag cement or portland-pozzolan cement conforming to the requirements of M 240, except that the pozzolan constituent in the Type IP portland pozzolan cement shall be fly ash and shall not exceed 25 percent by mass.
- 6.2.2. *Fly Ash*—Fly ash shall conform to the requirements of Class F or Class C of M 295.
- 6.2.3. *Allowable Combinations of Cementitious Materials*—the combination of cementitious materials used in the concrete shall be one of the following:
- 6.2.3.1. Portland cement only,
- 6.2.3.2. Portland blast-furnace slag cement only,

- 6.2.3.3. Portland-cement only, or
- 6.2.3.4. A combination of portland cement and fly ash wherein the proportion of fly ash is between 5 and 25 percent by mass of total cementitious material (portland cement plus fly ash).
- 6.3. *Aggregates*—*Aggregates* shall conform to M6 and M80, except that the requirement for
- 6.4. *Admixtures and blends* may be used with the approval of the owner.
- 6.5. *Steel reinforcement* Reinforcement shall consist of wire conforming to A325/A325M or A432/A432M, or of wire fabric conforming to A655/A655M or A675/A675M, or of bars of minimum Grade 300 steel conforming to A615/A615M.
- 6.6. *Synthetic fibers*—Collated fibrillated virgin polypropylene fibers may be used, at the owner's option, in concrete pipe as a nonstructural manufacturing material. Only Type III synthetic fibers designed and manufactured specifically for use in concrete and conforming to the requirements of ASTM Specification C 1116 shall be accepted.

7. DESIGN

- 7.1. *Design Table*—The diameter, wall thickness, compressive strength of the concrete, and the area of the circumferential reinforcement shall be as prescribed for Class I to V in Tables 1 to 5, except as provided in Section 7.2.
- 7.1.1. Footnotes to the tables herein are amended to be amplification of tabulated requirements and are to be considered applicable and binding as if they were contained in the body of the specification.
- 7.2. *Modified and Special Designs.*
- 7.2.1. If permitted by the owner the manufacturer may request approval of the owner of modified designs that differ from the designs in Section 7.1; or special designs for sizes and loads beyond those shown in Tables 1 to 5, Section 7.1, or special designs for pipe sizes that do not have steel reinforcement areas shown in Tables 1 to 5 of Section 7.1.
- 7.2.2. Such modified or special designs shall be based on rational or empirical evaluations of the ultimate strength and cracking behavior of the pipe and shall fully describe to the owner any deviations from the requirements of Section 7.1. The descriptions of modified or special designs shall include the wall thickness, the concrete strength, and the area, type, placement, number of layers, and strength of the steel reinforcement.
- 7.2.3.** The manufacturer shall submit to the owner proof of the adequacy of the proposed modified or special design. Such proof may comprise the submission of certified three-edge-bearing tests already made, which are acceptable to the owner or, if such three-edge-bearing tests are not available or acceptable, the manufacturer may be required to perform proof tests on sizes and classes selected by the owner to demonstrate the adequacy of the proposed design.
- 7.2.4. Such pipe must meet all of the test and performance requirements specified by the owner in accordance with Section 11.

- 7.3. A this specification, when the word "area" is not described by adjectives, such as cross-section or single wire, it shall be understood by the term "Actional area of reinforcement per unit lengths of pipe."

8 REINFORCEMENT

- 8.1. *Circumferential Reinforcement*—A line of circumferential reinforcement for any given total area may be composed of two layers for pipe with wall thicknesses of less than 180 mm or three layers for pipe with wall thicknesses of 180 mm or greater. The layers shall not be separated by more than the thickness of one longitudinal plus 6 mm. The multiple layers shall be fastened together to form a single cage. All other specification requirements such as laps, welds, and tolerances of placement in the wall of the pipe, etc. shall apply to this method of fabricating a line of reinforcement.
- 8.1.1. Where one line of circular reinforcement is used, it shall be placed from 33 to 30 percent of the wall thickness from the inner surface of the pipe, except that for wall thicknesses less than 63 mm, the protective cover of the concrete over the circumferential reinforcement in the wall of the pipe shall be 19 mm.
- 8.1.2. In pipe having two lines of circular reinforcement, each line shall be so placed that the protective covering of concrete over the circumferential reinforcement in the wall of the pipe shall be 25 mm.
- 8.1.3. In pipe having elliptical reinforcement with wall thicknesses 63 mm or greater, the reinforcement in the wall of the pipe shall be so placed that the protective covering of concrete over the circumferential reinforcement shall be 25 mm from the inner surface of the pipe at the vertical diameter and 23 mm from the outer surface of the pipe at the horizontal diameter. In pipe having elliptical reinforcement with wall thicknesses less than 63 mm, the protective covering of the concrete shall be 19 mm at the vertical and horizontal diameters.**
- 8.1.4. The location of the reinforcement shall be subject to the permissible variations in dimensions given in Section 12.5.
- 8.1.5. The spacing center-to-center of circumferential reinforcement in a cage shall not exceed 100 mm for pipe up to and including pipe having a 100-mm wall thickness nor exceed the wall thickness for larger pipe, and shall in no case exceed 150 mm.
- 8.1.6. Where the wall reinforcement does not extend into the joint, the maximum longitudinal distance to the last circumferential from the inside shoulder of the bell or the shoulder of the spigot shall be 75 mm, except that if this distance exceeds one-half the wall thickness, the pipe wall shall contain at least a total reinforcement area of the minimum specified area per linear meter times the length of the pipe section. The minimum cover on the last circumferential near the spigot shoulder shall be 13 mm.
- 8.1.6.1. Where reinforcement is in the bell or spigot the minimum end cover on the last circumferential shall be 13 mm in the bell or 6 mm in the spigot.**
- 8.1.7. The continuity of the circumferential reinforcing steel shall not be destroyed during the manufacture of the pipe, except that when agreed upon by the owner, lift eyes or holes may be provided in each pipe for the purpose of handling.
- 8.1.8. If splices are not welded, the reinforcement cut shall be lapped not less than 20 diameters for deformed bars and deformed cold-worked wire, and 40 diameters for plain bars and cold-drawn

in addition, where lapped cages of welded-wire fabric are used without welding, the lap shall

- 1.8.1.** ten split arc welded and arc not lapped to the minimum requirements above, pull tests of representative specimens shall develop at least 50 pct of the minimum specified strength of fire steel, and there shall be a minimum lap of 50 For butt-welded splices in bars or wire, permitted only with helically wound cages, pull tests of representative specimens shall develop at least 75 percent of the minimum specified strength of the steel.
- 2.** *Longitudinal Reinforcement*—Each line of circumferential reinforcement shall be assembled into a cage that shall contain sufficient longitudinal bars or members to maintain the reinforcement in shape and in position within the form to comply with permissible variations in Section 8.1. The exposure of the ends of longitudinals, stirrups, or spacers that have been used to position the cages during the placement of the concrete shall not be a cause for rejection.
- 3.** ***Joint Reinforcement*—The length of the joint as used herein means the inside length of the bell or the outside length of the spigot from the shoulder to the end of the pipe section. The end distances or cover on the end circumferential shall apply to any point on the circumference of the pipe or joint. When convoluted reinforcement is used, these distance and reinforcement areas shall be taken from the point on the convolutions closest to the end of the pipe section. Unless otherwise permitted by the owner, the following requirements for joint reinforcement shall apply.**
- 3.1.** *Joint Reinforcement for Non-Rubber Gasket Joints:*
- 3.1.1.** For pipe 900 mm and larger in diameter, either the bell or spigot shall contain circumferential reinforcement. This reinforcement shall be an extension of a wall cage, or may be a separate cage of at least the area per meter of that specified for the outer cage or one-half of that specified for single cage wall reinforcement, whichever is less.
- 3.1.2.** Where bells or spigots require reinforcement, the maximum end cover on the last circumferential shall be one-half the length of the joint or 75 mm, whichever is less.
- 3.2.** *Joint Reinforcement for Rubber Gasket Joints:*
- 3.2.1.** For pipe 300 mm and larger in diameter, the bell ends shall contain circumferential reinforcement. This reinforcement shall be an extension of the outer cage or a single wall cage, whichever is less, or may be a separate cage of at least the same area per meter with longitudinals as required in Section 8J. If a separate cage is used, the cage shall extend into the pipe with the last circumferential wire at least 25 mm past the inside shoulder where the pipe barrel meets the bell of the joint.
- 3.2.2.** Where bells require reinforcement, the maximum end cover on the last circumferential shall be 38 mm

JOINTS

- 1.** The joints shall be of such design and the ends of the concrete pipe sections so formed that when the sections are laid together they will make a continuous line of pipe with a smooth interior free from appreciable irregularities in the flow line, all compatible with the permissible variations even in Section 12.

J0. MANUFACTURE

- 10.1. **Water** The aggregates shall be sized, graded, proportioned, and mixed with such proportions of cementitious materials and water as will produce a homogeneous mass of such Quality that the pipe will **conform to** the test and design requirements of this specification. All concrete shall have a water-cementitious materials ratio not exceeding 0.43 by mass. Cementitious materials shall be as specified in Section 6J and shall be added to the mix in a proportion not less than 280 kg/m³ unless mix designs with a loss of cementitious materials content demonstrate that the quality and performance of the pipe meet the requirements of this specification.
- 10.2. **Curing** Pipe shall be subjected to any one of the methods of curing described in Sections 10.2.1 to 10.2.4 or to any other method or combination of methods approved by the owner, that will give satisfactory results. The pipe shall be cured for a sufficient length of time so that the specified D-load is obtained when acceptance is based on Section 5.1.1 or so that the concrete will develop the specified compressive strength at 28 days or less when acceptance is based on Section 5.1.2.
- 10.2.1. *Steam Curing*—Pipe may be placed in a curing chamber, free of outside drafts, and cured in a moist atmosphere maintained by the injection of steam for such time and such temperature as may be needed to enable the pipe to meet the strength requirements. The curing chamber shall be so constructed as to allow full circulation of steam around the entire pipe.
- 10.2.2. *Water Curing*—Concrete pipe may be water cured by covering with water-saturated material or by **a system of perforated pipes, zinc mechanical spacers, power hose, or by any other approved method that will keep the pipe moist during the specified curing period.**
- 10.2.3. The manufacturer has the option of combining the methods described in Sections 10.2.1 to 10.2.4, provided the required concrete compressive strength is attained.
- 10.2.4. A sealing membrane conforming to the requirements of M 148 may be applied and should be left intact until the required strength requirements are met. The concrete at the time of application shall be within 6°C of the atmospheric temperature. All surfaces shall be kept moist prior to the application of the compounds and shall be damp when the compound is applied.

11. PHYSICAL REQUIREMENTS

- 11.1. *Test Specimens*—The specified number of pipe required (or the tests shall be furnished without charge by the manufacturer and shall be selected at random by the owner, and shall be pipe that would not otherwise be rejected under this specification. The selection shall be at a point or points designated by the owner when placing the order. Unless designated otherwise by the owner in the order, the manufacturer shall select the test (external load crushing strength test, compression test of cylinders or compression test of cores) to be used to determine the acceptability of the pipe strength. Pipe that fails to meet the strength requirements as determined by the selected test type may not be rejected using one of the other test types without approval of the owner.
- 11.2. *Number and Type of Tests Required for Port of Delivery Schedules.*
- 11.2.1. *Preliminary Tests for Extended Delivery After*—An owner of pipe, whose needs require shipments at intervals over extended periods of time, shall be entitled to such tests, preliminary to delivery of pipe, as are required by the type of basis of acceptance specified by the owner in flow 1, of not more than three sections of pipe covering each time in which the owner is interested.

11.2.2. *Addition of Tests*—#Rev the preliminary tests described in Section 11.2.1, an owner may be required to add additional tests at such times as the owner may deem necessary provided that the total number of pipe tested (including preliminary tests) shall not exceed 1 pipe or 1 percent, whichever is the greater, of each size of pipe delivered

11.3. *External Load-Crushing Strength.*

11.3.1. The load to produce a 0.3-mm crack or the ultimate load, as determined by the three-edge-bearing method as described in T 280, shall be not less than that prescribed in Tables 1 to 5 for each respective class of pipe. Pipe that support the prescribed load to produce the 0.3-mm crack and do not show a wider crack shall be considered to have met the test requirement. It is not a requirement of this specification that the pipe be cracked or loaded to failure during these tests. Pipe that have been tested only to the formation of a 0.3-mm crack or lesser crack and that meet the 0.3-mm crack load requirements shall be accepted for use. Three-edge-bearing test to ultimate load is not required for any class of pipe 1500 mm or less in diameter listed in Tables 1-5 provided all other requirements of this specification are met.

Note 3—As used in this specification, the 0.3-mm crack is a test criterion for pipe tested in the three-edge-bearing test and is not intended as an indication of overstressed or failed pipe under installed conditions.

11.3.2. *Tests of Pipe to Determine External Load-Crushing Strength Requirements*—Pipe shall be considered as meeting the strength requirements when all test specimens conform to the strength requirements. Should any of the test specimens fail to meet the strength requirements, the manufacturer shall be allowed a retest on two additional specimens for each specimen that failed, and the pipe shall be acceptable only when all of the retest specimens meet the strength requirements.

11.4. *Type of Specimen*—Compression tests determining concrete compressive strength may be made on either standard rod-cast concrete cylinders or concrete cylinders compacted and cured in like manner as the pipe, or on cores drilled from the pipe.

11.5. *Compression Test of Cylinders:*

11.5.1. *Cylinder Preparation*—Cylinders shall be prepared in accordance with Section 11 of T 280.

11.5.2. *Number of Cylinders*—Prepare no fewer than five test cylinders from a group (one day's production of each concrete strength) of pipe sections.

11.5.3. *Acceptance on the Basis of Cylinder Test Results:*

11.5.3.1. When the compressive strengths of all cylinders tested for a group are equal to or greater than the required concrete strength, the compressive strength of concrete in the group of pipe sections shall be accepted.

11.5.3.2. When the average compressive strength of all cylinders tested is equal to or greater than the required concrete strength, and not more than 10 percent of the cylinders tested have a compressive strength less than the required concrete strength, and no cylinder tested has a compressive strength less than 80 percent of the required concrete strength, then the group shall be accepted.

11.5.3.3. The pipe shall be acceptable only when the compressive strength of the cylinders tested conforms to the acceptance criteria stated in Section 11.5.3.1 or Section 11.5.3.2.

- 11.6. *Compression Testing of Cores:*
- 11.6.J. *Obtaining Gages:* Cores shall be obtained and prepared in accordance with Section 6 of T 280.
- 11.6.2. *Number of Cores:* One core shall be taken from a pipe section selected at random from each day's production run of a single concrete strength.
- 11.7. *Acceptance on the Basis of Core Test Results:*
- 11.7.1. When the compressive strength of a core tested (or a group of pipe sections) is equal to or greater than the required concrete strength, the compressive strength of the concrete for the group is acceptable.
- 11.7.2. If the compressive strength of the core tested is less than the required concrete strength, two additional cores shall be taken from that pipe section and tested. Concrete represented by these core tests shall be considered acceptable if (1) the average of three cores is equal to at least 85 percent of the required strength, and (2) no single core is less than 75 percent of the required**
- 11.7.3. If the compressive strength of the three cores does not meet the requirements of Section 11.7.2, the pipe section from which the core was taken shall be rejected. Two pipe sections from the remainder of the group shall be selected at random and cored and tested for conformance with either Section 11.7.1 or Section 11.7.2. If both pipe sections meet the core strength requirements of Section 11.7.1 or Section 11.7.2, the remainder of the group shall be acceptable. If both pipe sections do not meet the test strength requirement, the remainder of the group shall be either rejected or, at the option of the manufacturer, each pipe section of the remaining group shall be tested and accepted individually and any of the pipe sections that have core strengths less than the requirements of Section 11.7.1 or Section 11.7.2 shall be rejected.**
- 11.8. *Plugging Core Holes:* Core holes shall be plugged and sealed by the manufacturer in a manner such that the pipe section will meet all of the requirements of this specification. Pipe sections so plugged and sealed shall be considered satisfactory for use.
- 11.9. Absorption: The absorption of a sample from the wall of the pipe, as determined in accordance with T 280, shall not exceed 9 percent of the dry mass. Each sample shall have a minimum mass of 1.0 kg, shall be free of visible cracks, and shall represent the full wall thickness of the pipe. When the initial absorption sample from a pipe fails to conform to this specification, the absorption test shall be made on another sample from the same pipe and the results of the retest shall be substituted for the original test results.**
- 11.10. Rejection of Pipe: When not more than 20 percent of the concrete specimens fail to pass the requirements of this specification, the manufacturer may cull the project stock and may eliminate whatever quantity of pipe desired and shall mark those pipes so that they will not be shipped. The required tests shall be made on the balance of the order and the pipes shall be accepted if they conform to the requirements of this specification.**
- 11.11. *Test Equipment:* Any manufacturer furnishing pipe under this specification shall furnish all facilities and personnel necessary to carry out the tests described in T 280.

PERMISSIBLE VARIATIONS

Internal Diameter—See Table 6. At the manufacturer's option the design diameter shall be the designated diameter of the convuluted English diameter. The internal diameter of 300 to 600 mm pipe shall vary not more than $\pm 1\%$ percent from the design diameter. The internal diameter for pipe larger than 600 mm shall not vary from the design diameter by more than 11 percent of the design diameter or a 10 mm, whichever is greater. Pipe sections that are intended to be jointed to each other shall be furnished with the same design diameter.

Wall Thickness—The wall thickness shall not vary more than shown in the design or specified wall by more than $\pm 1\%$ percent or 5 mm, whichever is greater. A specified wall thickness more than required in the design is not cause for rejection. Pipe having localized variations in wall thickness exceeding those specified above shall be accepted if the three-edge-bearing strength and minimum steel cover requirements are met.

3. *Length of Two Opposite Sides*—Variations in the laying length of two opposite sides of the pipe shall not be more than 6 mm for all sizes through 600-mm internal diameter, and not more than 10 mm/m of internal diameter for all sizes larger with a maximum of 16 mm in any length of pipe through 2100-mm internal diameter, and a maximum of 19 mm for 2250mm internal diameter or larger, except where beveled end pipe for laying on curves is specified by the owner.
4. *Length of Pipe* The variation in length of a section of pipe shall not be more than 10 mm/m with a maximum of 13 mm in any length of pipe. Regardless of the variation or overage in any section of pipe, the end cover requirements of Sections 8 and 12 shall apply.
5. *Position or Area of Reinforcement:*
 - 5.1. *Position*—The maximum variation in the position of the reinforcement shall be $\pm 10\%$ percent of the wall thickness or ± 13 mm, whichever is greater. Pipe having variations in the position of the reinforcement exceeding those specified above shall be accepted if the three-edge-bearing strength requirements obtained on a representative specimen are met. In no case, however, shall the cover over the circumferential reinforcement be less than 6 mm as measured to the end of the spigot or 13 mm as measured to any other surface. The preceding minimum cover limitations do not apply to mating surfaces of non-rubber gasket joints or gasket grooves in rubber gasket joints. If convoluted reinforcement is used, the convoluted circumferential end wire may be at the end surface of the joint providing the alternate convolutions have at least 25 mm cover from the end surface of the joint.
 - 5.2. *Area of Reinforcement*—Reinforcement will be considered as meeting the design requirements if the area, computed on the basis of nominal area of the wire or bars used, equals or exceeds the requirements of Section 7.1 or Section 7.2. Actual area of the reinforcing used may vary from the nominal area according to permissible variations of the standard specifications for the reinforcing. When inner cage and outer cage reinforcing is used, the inner cage nominal area may vary to the lower limit of 85 percent of the elliptical nominal area and the outer cage nominal area may vary to the lower limit of 51 percent of the elliptical nominal area provided that the total nominal area of the inner cage plus the outer cage shall not vary beyond the lower limit of 140 percent of the elliptical nominal area.

Table Permissible Variation in Internal Diameter

Designated Diameter of	Equivaleni English	pqq	glish
Pipe, mm	Dia, in.	Dia, mm	
300	12	30S	
375	1\$	3gl	
450	18	457	
123	21	533	
600	24	610	
675	27	686	
750	30	762	
825	33	838	
900	36	914	
1050	42	1067	
1200	48	1219	
1350	S4	1372	
1500	60	1524	
1659	66	1676	
1S00	72	1829	
i950	78	1981	
2100	84	2134	
2250	90	2286	
2400	9d	2438	
2550	102	2591	
2700	108	2743	
2830	114	2896	
3000	120	iM8	
3150	126	3200	
3300	132	3333	
3450	U38	3S05	
ice	i<i	3ass	

13. REPAIRS

13.1. Pipe may be repaired, if necessary, because of imperfections in manufacturing or damage during handling and will be acceptable if, in the opinion of the owner, the repaired pipt conforms io the requirements of this specification.

14. INSPECTION

14.1. The quality of materials, the process of manufacture, a»d a« rashcd pipe shall be subject to inspection and approval by the owner.

15. REJECTION

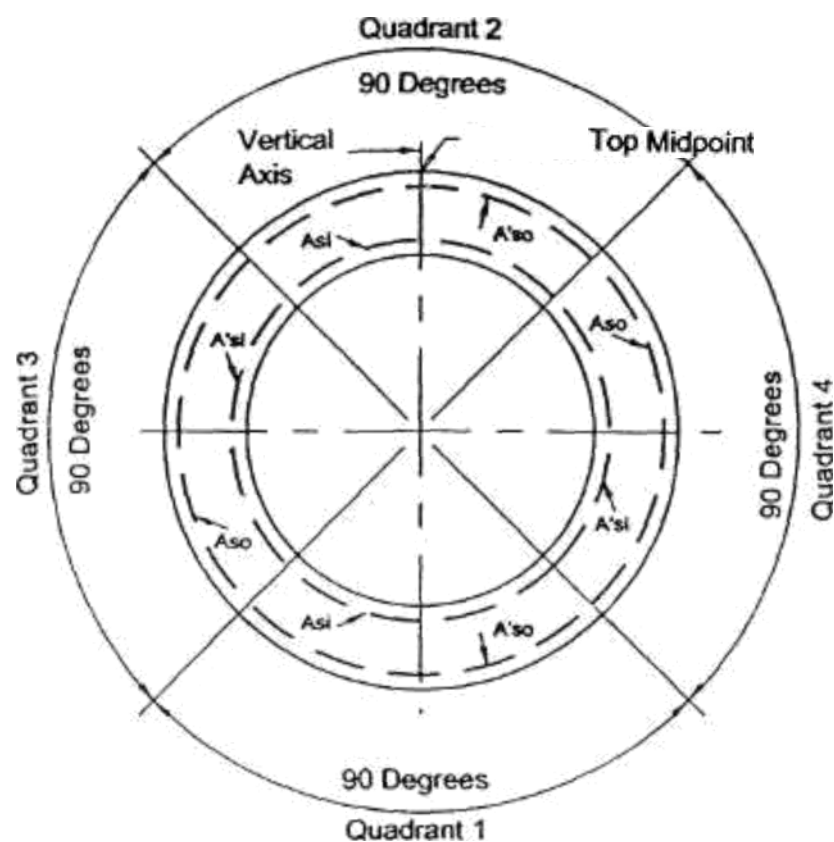
15.1. Pipe shall be subject to rejection if it fails to conform to any of the specification requirements. Individual sections of pipe may be rejected because of any of the following:

15.1.1. Fractures or cracks passing through the wall, except for a single end crack that does not exceed the depth of the joint.

- 15.1.2.** Defects that indicate proportioning, mixing, and curing not in compliance with Section 10.1 or surface defects indicating honey-comb or pitting texture that would adversely affect the function of the pipe.
- 15.1.3. The ends of the pipe are not normal to the walls and centerline of the pipe, within the limits of variations given in Sections 12.3 and 12.4.
- 15.1.4.** Damaged or cracked ends where such damage would prevent making a satisfactory joint.
- 15.1.5. Any continuous crack having a surface width of 0.3 mm or more and extending for a length of 300 mm or more, regardless of position in the wall of the pipe.

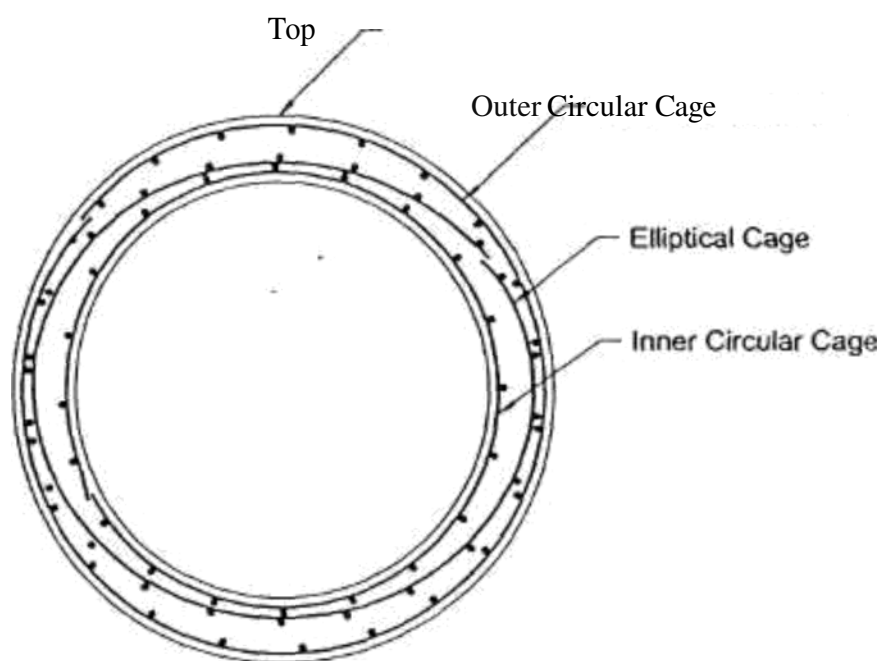
16. PRODUCT MARKING

- 16.1. The following information shall be clearly marked on each section of pipe:
- 16.1.1.** The pipe class and specification designation,
- 16.1.2. The date of manufacture,
- 16.1.3. The name or trademark of the manufacturer, and
- 16.1.4. Identification number.
- 16.2. One end of each section of pipe with elliptical or quadrant reinforcement shall be clearly marked during the process of manufacturing or immediately thereafter, on the inside and the outside of opposite walls along the minor axis of the elliptical reinforcing or along the vertical axis for quadrant reinforcing.
- 16.3.** Markings shall be indented on the pipe section or painted thereon with waterproof paint.



- Notes:
1. The total reinforcement area (A'si) of the inner cage plus the quadrant material in Quadrants 1 and 2 shall not be less than that specified for the inner cage in Tables 1 to 5.
 2. The total reinforcement area (A'so) of the outer cage plus the quadrant material in Quadrants 3 and 4 shall not be less than that specified for the outer cage in Tables 1 to 5.
 3. The total reinforcement area (A'si) of the inner cage in quadrants 3 and 4 shall not be less than 5 percent of that specified for the inner cage in Tables 1 to 3.
 4. The total reinforcement area (A'so) of the outer cage in Quadrants 1 and 2 shall not be less than 5 percent of that specified for the outer cage in Tables 1 to 3.
 5. If the reinforcement area (A'so) of the outer cage in Quadrants 1 and 2 is less than 50 percent of that specified for the outer cage in Tables 1 to 5, the quadrant material used for the outer cage in Quadrants 3 and 4 shall extend a minimum of 4 inches from the quadrant material used for the outer cage in Quadrants 1 and 2 to a distance equal to the wall thickness as specified in Tables 1 to 5.

Figure 1—Quadrant Reinforcement



- Notes:
1. The total reinforcement area of the inner circular cage and the elliptical cage shall not be less than that specified for the inner cage in Tables 1 to 3.
 2. The total reinforcement area of the outer circular cage and the elliptical cage shall not be less than that specified for the outer cage in Tables 1 to 5.

Figure 2—Triple Cage Reinforcement

17. KEYWORDS

17.1. Circular pipe; culvert; D-load; pipe; reinforced concrete; sewer pipe: atorm drain.

Agrees with ASTM C 76bI-02 except for limiting the absorption testing to the five-hour boil method, the use of synthetic fibers are at the option of the owner, fly ash limitation in Section 6.2d.4 and the mcthcd of acceptance is changed as per Sections 1 1.1, 1 1.5.3.3, and l 1.7.3.